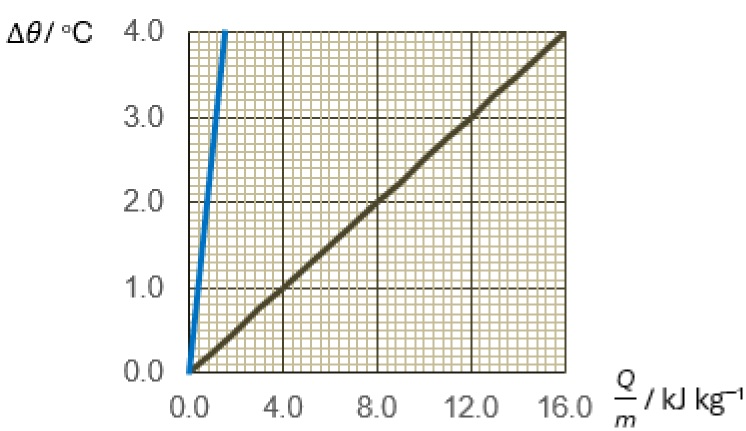
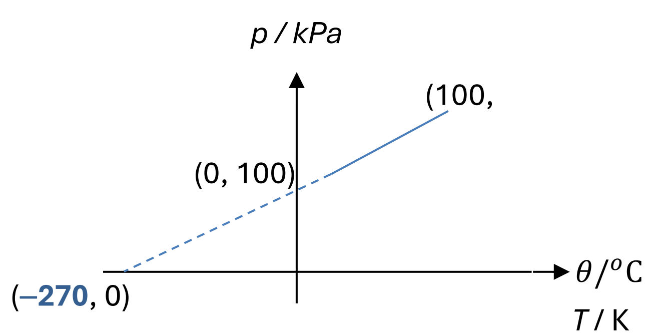
Questions			Answers	Marks
9	(a)	(i)	No net heat transfer between A and B	B1
			A and B have same temperature	B1
		(ii)	gradient of graph = $\frac{1}{c}$	M1
			$c = \frac{1}{\text{gradient}} = \frac{1}{4/16} = 4\ \text{kJ kg}^{-1}\ \text{K}^{-1}$	A1
		(iii)	$m_A c_A \Delta\theta_A = m_B c_B \Delta\theta_B$ gives $c_A = \frac{1.5}{5.0} \times \frac{20}{60} \times 4000$	M1
			$c_A = 400\ \text{J kg}^{-1}\ \text{K}^{-1}$	A1
		(iv)	 straight line through origin, 10 times as steep.	B1
	(b)	(i)	Δu : increase in internal energy q : heat supplied to the system w : work done on the system M1: definition of quantity A1: direction of change (underlined words)	M1 A1

	Δu	q	w
process 1 (adiabatic)	positive	zero	positive
process 2 (constant volume)	negative	negative	zero
process 3 (isothermal)	zero	positive	negative

		(ii)	first row correct	B1
		(iii)	second row correct	B1
			Volume remains unchanged (hence no work done)	B1
			Since heat is lost to the surrounding to melt ice-water mixture, either • $q < 0$, and $\boldsymbol{\Delta u = q + 0}$, so $\Delta u < 0$. or • temperature decreases <u>and</u> internal energy , $u \propto T$ thermodynamic temperature, so $\Delta u < 0$.	B1
		(iv)	third row correct	B1
			• gas expands, so $w < 0$ <u>and</u> at • constant temperature so $\Delta u = 0$	B1
		(v)	for the cycle $\Delta u = 0$, so $\boldsymbol{0 = q + w}$	B1
			• $w = -q = -(-100 \times 334) = +3.34 \times 10^4$ J	A1
	(c)	**(i)**	 	M1
			Gradient = $\frac{0.37}{100} = 0.37$, so $y = 0.37x + 100$	A1
			When $y = 0,\ x = -\frac{100}{0.37} = -270$ or	

			Similar triangles give $\frac{\lvert\theta\rvert}{100} = \frac{\lvert\theta\rvert+100}{137}$ gives $\lvert\theta\rvert = 270$ so $\theta = -270$ $\frac{100}{T-(-273.15)} = \frac{37}{100}$, $T = -2.88$, so $\theta = -273.15 - (-2.88) = -270$	
		(ii)	• at low pressure gases are <u>far apart</u> and seldom interact with one another • so intermolecular forces related to distance between molecules) are negligible.	B1